
Bayesian Online Changepoint Detection

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Abstract

Changepoints are abrupt variations in the generative parameters of a data sequence. Online detection of changepoints is useful in modelling and prediction of time series in application areas such as finance, biometrics, and robotics. While frequentist methods have yielded online filtering and prediction techniques, most Bayesian papers have focused on the retrospective segmentation problem. Here we examine the case where the model parameters before and after the changepoint are independent and we derive an online algorithm for exact inference of the most recent changepoint. We compute the probability distribution of the length of the current “run,” or time since the last changepoint, using a simple message-passing algorithm. Our implementation is highly modular so that the algorithm may be applied to a variety of types of data. We illustrate this modularity by demonstrating the algorithm on three different real-world data sets.

1 INTRODUCTION

Changepoint detection is the identification of abrupt changes in the generative parameters of sequential data. As an online and offline signal processing tool, it has proven to be useful in applications such as process control [1], EEG analysis [5, 2, 17], DNA segmentation [6], econometrics [7, 18], and disease demographics [9].

Frequentist approaches to changepoint detection, from the pioneering work of Page [22, 23] and Lorden [19] to recent work using support vector machines [10], offer online changepoint detectors. Most Bayesian approaches to changepoint detection, in contrast, have been offline and retrospective [24, 4, 26, 13, 8]. With a

few exceptions [16, 20], the Bayesian papers on changepoint detection focus on segmentation and techniques to generate samples from the posterior distribution over changepoint locations.

In this paper, we present a Bayesian changepoint detection algorithm for online inference. Rather than retrospective segmentation, we focus on causal predictive filtering; generating an accurate distribution of the next unseen datum in the sequence, given only data already observed. For many applications in machine intelligence, this is a natural requirement. Robots must navigate based on past sensor data from an environment that may have abruptly changed: a door may be closed now, for example, or the furniture may have been moved. In vision systems, the brightness change when a light switch is flipped or when the sun comes out.

We assume that a sequence of observations $x_1, x_2, \dots, x_T$ may be divided into non-overlapping *product partitions* [3]. The delineations between partitions are called the changepoints. We further assume that for each partition ρ , the data within it are i.i.d. from some probability distribution $P(x_t | \boldsymbol{\eta}_\rho)$. The parameters $\boldsymbol{\eta}_\rho$, $\rho = 1, 2, \dots$ are taken to be i.i.d. as well. We denote the contiguous set of observations between time a and b inclusive as $\mathbf{x}_{a:b}$. The discrete *a priori* probability distribution over the interval between changepoints is denoted as $P_{\text{gap}}(g)$.

We are concerned with estimating the posterior distribution over the current “run length,” or time since the last changepoint, given the data so far observed. We denote the length of the current run at time t by r_t . We also use the notation $\mathbf{x}_t^{(r)}$ to indicate the set of observations associated with the run r_t . As r may be zero, the set $\mathbf{x}^{(r)}$ may be empty. We illustrate the relationship between the run length r and some hypothetical univariate data in Figures 1(a) and 1(b).

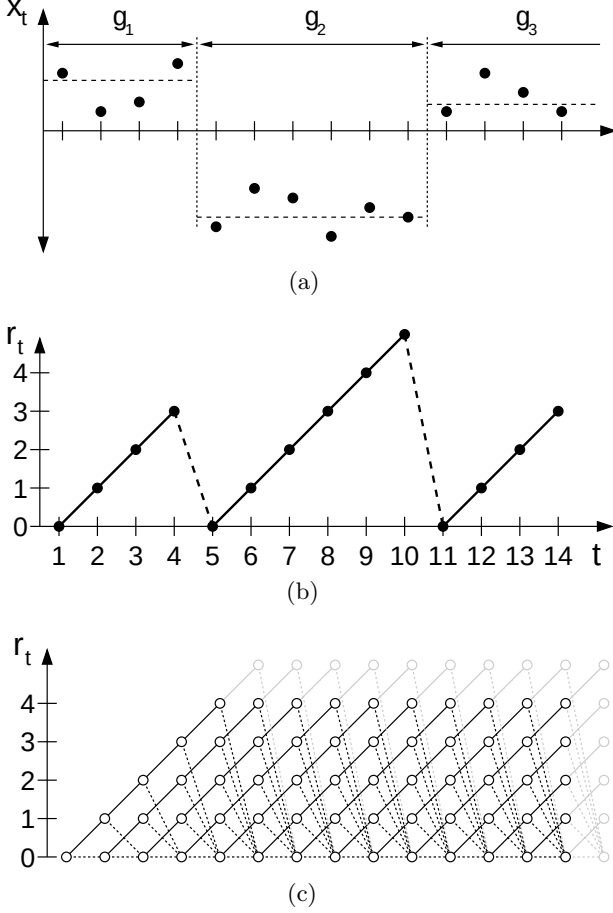


Figure 1: This figure illustrates how we describe a changepoint model expressed in terms of run lengths. Figure 1(a) shows hypothetical univariate data divided by changepoints on the mean into three segments of lengths $g_1 = 4$, $g_2 = 6$, and an undetermined length g_3 . Figure 1(b) shows the run length r_t as a function of time. r_t drops to zero when a changepoint occurs. Figure 1(c) shows the trellis on which the message-passing algorithm lives. Solid lines indicate that probability mass is being passed “upwards,” causing the run length to grow at the next time step. Dotted lines indicate the possibility that the current run is truncated and the run length drops to zero.

2 RECURSIVE RUN LENGTH ESTIMATION

We assume that we can compute the predictive distribution conditional on a given run length r_t . We then integrate over the posterior distribution on the current run length to find the marginal predictive distribution:

$$P(x_{t+1} | \mathbf{x}_{1:t}) = \sum_{r_t} P(x_{t+1} | r_t, \mathbf{x}_t^{(r)}) P(r_t | \mathbf{x}_{1:t}) \quad (1)$$

To find the posterior distribution

$$P(r_t | \mathbf{x}_{1:t}) = \frac{P(r_t, \mathbf{x}_{1:t})}{P(\mathbf{x}_{1:t})}, \quad (2)$$

we write the joint distribution over run length and observed data recursively.

$$\begin{aligned} P(r_t, \mathbf{x}_{1:t}) &= \sum_{r_{t-1}} P(r_t, r_{t-1}, \mathbf{x}_{1:t}) \\ &= \sum_{r_{t-1}} P(r_t, x_t | r_{t-1}, \mathbf{x}_{1:t-1}) P(r_{t-1}, \mathbf{x}_{1:t-1}) \quad (3) \\ &= \sum_{r_{t-1}} P(r_t | r_{t-1}) P(x_t | r_{t-1}, \mathbf{x}_t^{(r)}) P(r_{t-1}, \mathbf{x}_{1:t-1}) \end{aligned}$$

Note that the predictive distribution $P(x_t | r_{t-1}, \mathbf{x}_{1:t})$ depends only on the recent data $\mathbf{x}_t^{(r)}$. We can thus generate a recursive message-passing algorithm for the joint distribution over the current run length and the data, based on two calculations: 1) the prior over r_t given r_{t-1} , and 2) the predictive distribution over the newly-observed datum, given the data since the last changepoint.

2.1 THE CHANGEPPOINT PRIOR

The conditional prior on the changepoint $P(r_t | r_{t-1})$ gives this algorithm its computational efficiency, as it has nonzero mass at only two outcomes: the run length either continues to grow and $r_t = r_{t-1} + 1$ or a changepoint occurs and $r_t = 0$.

$$P(r_t | r_{t-1}) = \begin{cases} H(r_{t-1} + 1) & \text{if } r_t = 0 \\ 1 - H(r_{t-1} + 1) & \text{if } r_t = r_{t-1} + 1 \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

The function $H(\tau)$ is the *hazard function*. [11].

$$H(\tau) = \frac{P_{\text{gap}}(g=\tau)}{\sum_{t=\tau}^{\infty} P_{\text{gap}}(g=t)} \quad (5)$$

In the special case is where $P_{\text{gap}}(g)$ is a discrete exponential (geometric) distribution with timescale λ , the process is memoryless and the hazard function is constant at $H(\tau) = 1/\lambda$.

Figure 1(c) illustrates the resulting message-passing algorithm. In this diagram, the circles represent run-length hypotheses. The lines between the circles show recursive transfer of mass between time steps. Solid lines indicate that probability mass is being passed “upwards,” causing the run length to grow at the next time step. Dotted lines indicate that the current run is truncated and the run length drops to zero.

2.2 BOUNDARY CONDITIONS

A recursive algorithm must not only define the recurrence relation, but also the initialization conditions. We consider two cases: 1) a changepoint occurred *a priori* before the first datum, such as when observing a game. In such cases we place all of the probability mass for the initial run length at zero, i.e. $P(r_0=0) = 1$. 2) We observe some recent subset of the data, such as when modelling climate change. In this case the prior over the initial run length is the normalized *survival function* [11]

$$P(r_0=\tau) = \frac{1}{Z} S(\tau), \quad (6)$$

where Z is an appropriate normalizing constant, and

$$S(\tau) = \sum_{t=\tau+1}^{\infty} P_{\text{gap}}(g=t). \quad (7)$$

2.3 CONJUGATE-EXPONENTIAL MODELS

Conjugate-exponential models are particularly convenient for integrating with the changepoint detection scheme described here. Exponential family likelihoods allow inference with a finite number of sufficient statistics which can be calculated incrementally as data arrives. Exponential family likelihoods have the form

$$P(\mathbf{x} | \boldsymbol{\eta}) = h(\mathbf{x}) \exp(\boldsymbol{\eta}^\top \mathbf{U}(\mathbf{x}) - A(\boldsymbol{\eta})) \quad (8)$$

where

$$A(\boldsymbol{\eta}) = \log \int d\boldsymbol{\eta} h(\mathbf{x}) \exp(\boldsymbol{\eta}^\top \mathbf{U}(\mathbf{x})). \quad (9)$$

The strength of the conjugate-exponential representation is that both the prior and posterior take the form of an exponential-family distribution over $\boldsymbol{\eta}$ that can be summarized by succinct hyperparameters ν and χ .

$$P(\boldsymbol{\eta} | \chi, \nu) = \tilde{h}(\boldsymbol{\eta}) \exp(\boldsymbol{\eta}^\top \chi - \nu A(\boldsymbol{\eta}) - \tilde{A}(\chi, \nu)) \quad (10)$$

We wish to infer the parameter vector $\boldsymbol{\eta}$ associated with the data from a current run length r_t . We denote this run-specific model parameter as $\boldsymbol{\eta}_t^{(r)}$. After finding the posterior distribution $P(\boldsymbol{\eta}_t^{(r)} | r_t, \mathbf{x}_t^{(r)})$, we can marginalize out the parameters to find the predictive distribution, conditional on the length of the current run.

$$P(x_{t+1} | r_t) = \int d\boldsymbol{\eta} P(x_{t+1} | \boldsymbol{\eta}) P(\boldsymbol{\eta}_t^{(r)} = \boldsymbol{\eta} | r_t, \mathbf{x}_t^{(r)}) \quad (11)$$

1. Initialize

$$P(r_0) = \tilde{S}(r) \text{ or } P(r_0=0) = 1$$

$$\nu_1^{(0)} = \nu_{\text{prior}}$$

$$\chi_1^{(0)} = \chi_{\text{prior}}$$

2. Observe New Datum x_t

3. Evaluate Predictive Probability

$$\pi_t^{(r)} = P(x_t | \nu_t^{(r)}, \chi_t^{(r)})$$

4. Calculate Growth Probabilities

$$P(r_t=r_{t-1}+1, \mathbf{x}_{1:t}) = P(r_{t-1}, \mathbf{x}_{1:t-1}) \pi_t^{(r)} (1-H(r_{t-1}))$$

5. Calculate Changepoint Probabilities

$$P(r_t=0, \mathbf{x}_{1:t}) = \sum_{r_{t-1}} P(r_{t-1}, \mathbf{x}_{1:t-1}) \pi_t^{(r)} H(r_{t-1})$$

6. Calculate Evidence

$$P(\mathbf{x}_{1:t}) = \sum_{r_t} P(r_t, \mathbf{x}_{1:t})$$

7. Determine Run Length Distribution

$$P(r_t | \mathbf{x}_{1:t}) = P(r_t, \mathbf{x}_{1:t}) / P(\mathbf{x}_{1:t})$$

8. Update Sufficient Statistics

$$\nu_{t+1}^{(0)} = \nu_{\text{prior}}$$

$$\chi_{t+1}^{(0)} = \chi_{\text{prior}}$$

$$\nu_{t+1}^{(r+1)} = \nu_t^{(r)} + 1$$

$$\chi_{t+1}^{(r+1)} = \chi_t^{(r)} + \mathbf{u}(x_t)$$

9. Perform Prediction

$$P(x_{t+1} | \mathbf{x}_{1:t}) = \sum_{r_t} P(\mathbf{x}_{t+1} | \mathbf{x}_t^{(r)}, r_t) P(r_t | \mathbf{x}_{1:t})$$

10. Return to Step 2

Algorithm 1: The online changepoint algorithm with prediction. An additional optimization not shown is to truncate the per-timestep vectors when the tail of $P(r_t | \mathbf{x}_{1:t})$ has mass beneath a threshold.

This marginal predictive distribution, while generally not itself an exponential-family distribution, is usually a simple function of the sufficient statistics. When exact distributions are not available, compact approximations such as that described by Snelson and Ghahramani [25] may be useful. We will only address the exact case in this paper, where the predictive distribution associated with a particular current run length is parameterized by $\nu_t^{(r)}$ and $\chi_t^{(r)}$.

$$\nu_t^{(r)} = \nu_{\text{prior}} + r_t \quad (12)$$

$$\chi_t^{(r)} = \chi_{\text{prior}} + \sum_{t' \in r_t} \mathbf{u}(x_{t'}) \quad (13)$$